

Ares Koumblis

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EDUCATION

Carnegie Mellon University

Master of Science in Robotics

Pittsburgh, PA

Expected Aug 2028

National Technical University of Athens

Integrated Master's (BE/ME) in Electrical and Computer Engineering

Athens, Greece

Feb 2024

GPA: 9.3/10.0 (top 3%)

PROFESSIONAL EXPERIENCE

Tesla

Palo Alto, CA

Senior Autopilot Software Development Engineer

Aug 2025 – Jul 2026

Autopilot Software Development Engineer

Feb 2025 – Aug 2025

Autopilot Software QA Engineer

Jan 2024 – Feb 2025

- Fine-tuned end-to-end driving policies with GRPO, reducing speeding in low-visibility conditions without regressing ride comfort
- Enhanced explicit trajectory selection logic, enabling FSD v14 on HW3 vehicles and left-hand-traffic markets
- Implemented reduced-visibility slowdown logic across vehicle platforms, reducing collision rate in simulation by 44%
- Led validation of Tesla's closed-loop video-generation simulator, demonstrating strong correlation with both track-test results and real-world miles per critical intervention
- Developed safety and comfort evaluation metrics across Tesla's open-loop, closed-loop, and video-generation simulators, enabling automated rejection of underperforming models and reducing reliance on on-vehicle testing
- Adapted Lane Departure Avoidance for rear-wheel steering and deployed on Cybertruck for 5-star NCAP
- Led platform-specific validation and release efforts for the first FSD deployment on Cybertruck (~30,000 vehicles)

RESEARCH

Spatial Reasoning in 3D Vision-Language Models

Pittsburgh, PA

Carnegie Mellon University – Katerina Fragkiadaki Lab

Aug 2026 – Present

- Built chain-of-thought training data and added tool calling, multi-turn reasoning, and viewpoint reasoning to an in-house 3D vision-language model
- Extending it to recurse on zoomed views or new viewpoints, bounding context growth

PROJECTS

Autonomous Formula Student (FSAE) Race Car

Athens, Greece

National Technical University of Athens

Oct 2021 – Aug 2023

- Founded and led a 10-member team to develop and deliver an autonomous race car over a 2-year timeline
- Designed and deployed an edge perception stack (YOLOv5 on TPU, ResNet keypoint detector on iGPU) for real-time track cone detection and distance estimation, earning 4th place in the ZEISS Real-Time Video Processing Award
- Developed a YOLOv8 variant with a regression head for direct cone distance estimation via knowledge distillation from the baseline perception stack
- Built a graph-based SLAM system using the iSAM algorithm for real-time cone and vehicle localization
- Trained an encoder-decoder network to model nonlinear vehicle dynamics, deployed in a Koopman MPC for path tracking

SKILLS

Programming Languages: Python, C++, C

Machine Learning: PyTorch, HuggingFace, Scikit-Learn, OpenCV

Robotics: ROS2, GTSAM

Developer Tools: Git, Linux, Weights & Biases